EdgePilot Architecture & Metric Report

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WORKLOAD SPECIFICATIONS

Latency target: 100 ms | Budget: 500 INR/month | Privacy: medium

Criticality: medium | Bandwidth: medium

RECOMMENDED DEPLOYMENT TIER

CLOUD

1. Infrastructure Economics & Grounding Summary

- End-to-End Latency: 120 ms
- Estimated Monthly Operating Cost: INR 300 / month

Suitability scores / 100: on device: 58 | far edge: 70.3 | regional edge: 64.2 | cloud: 74.2

Scores are ranking indicators, not benchmark measurements, probabilities, or deployment guarantees.

2. Tier Comparison: Latency, Cost & Evidence

TIER	LATENCY	COST / MO.	LATENCY EVIDENCE	COST EVIDENCE	SLA / BUDGET
ON DEVICE	15 ms	INR 900	estimated: Qualcomm NPU Profile (Snapdragon)	estimated: Calibrated Device & Ops Estimate	Lat PROVISIONAL PASS Cost PROVISIONAL
FAR EDGE	30 ms	INR 500	estimated: Standard Network & Hosting Estimate	estimated: Calibrated Device & Ops Estimate	Lat PROVISIONAL PASS Cost PROVISIONAL
REGIONAL EDGE	55 ms	INR 1,100	estimated: Standard Network & Hosting Estimate	estimated: Calibrated Device & Ops Estimate	Lat PROVISIONAL PASS Cost PROVISIONAL
CLOUD	120 ms	INR 300	estimated: Standard Network & Hosting Estimate	estimated: Calibrated Device & Ops Estimate	Lat PROVISIONAL FAIL Cost PROVISIONAL

3. Detailed Component Breakdown (Selected Tier)

Latency Components (ms):

preprocessing: 5ms | upload: 22ms | network_rtt: 60ms | queue: 8ms | inference: 7ms | postprocessing: 2ms | download: 16ms

Cost Components (INR):

compute: INR 100 | network: INR 0 | operations: INR 200

Evidence & assumptions:

Selected-tier source type: latency estimated; cost estimated. Constraint checks compare supplied values to workload limits; checks based on estimates are provisional. Illustrative upload, queue, RTT, and return path durations; Illustrative, not a public cloud provider price quote

Recommendation rationale:

Cloud ranks first with a provisional, uncalibrated suitability score of 74.2/100. Cloud: end-to-end latency 120 ms (estimated); monthly cost INR 300 (estimated). Estimated latency exceeds the supplied limit; treat this recommendation as provisional until verified. Far edge ranks next (70.3/100). The ranking is not a guarantee of deployment performance.

Limitations: Estimated figures are provisional until validated with representative application measurements and workload-matched pricing. Verify the selected tier against production latency, throughput, memory, privacy, and criticality requirements before deployment.